Equations & Inequalities: Mini-Lessons

NC.8.EE.7

Name _____ Date _____ Period _____

3.1 One- and two-step equations NC.8.EE.7

To **solve** an equation, get **x alone**. Undo each move with its **opposite**, and do it to **both sides** so the equation stays balanced.

Variable
a letter, like x, that stands for a number

Inverse (opposite) operation
the move that undoes another: + and − undo each other, × and ÷ undo each other

Solution
the number that makes the equation true

x = 4

Check: 2(4) + 3 = 11 ✓



To undo...	+ 5	− 4	× 3 (3x)	÷ 2
do this	− 5	+ 4	÷ 3	× 2

Order: undo + or − first, then × or ÷.

Watch out: Do the same move to **both sides**. A move on one side only breaks the balance.

Walk through: `Solve 3x - 4 = 11`

1. Which move comes first?

_____

2. x is multiplied by 3. What undoes that?

_____

3. Check: put 5 in for x. What is 3(5) − 4?

_____

Check your understanding

1 Score 3 NC.8.EE.7

Solve: $x + 7 = 15$

Answer: _____

2 Score 3 NC.8.EE.7

What is the **first** move to solve $4x + 5 = 21$?

(A) Add 5 to both sides

(B) Divide both sides by 4

(C) Subtract 5 from both sides

(D) Subtract 4 from both sides

3 Score 3 NC.8.EE.7

Solve: $2x + 3 = 11$

Answer: _____

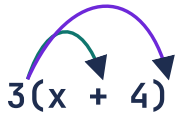
3.2 Distribute and combine like terms NC.8.EE.7

The number outside the parentheses **multiplies every term inside**. **Like terms** (x-terms with x-terms, numbers with numbers) can be added together.

Distribute
multiply the outside number by each term inside the ( )

Like terms
terms with the same variable part: 3x and 5x, or 4 and 9

Coefficient
the number in front of a variable (in 3x, the 3)



$3 \cdot x + 3 \cdot 4$

$3x + 12$

$4(x - 2) = 4x - 8$

$3x + 5 + 2x = 5x + 5$

Watch out: 3(x + 4) is **not** 3x + 4. The 3 multiplies the 4 too: 3x + 12.

Walk through: Solve $2(x + 5) = 16$

1 There are parentheses. What is the first move?

_____

2 Undo the + 10.

3 Get x alone.

Check your understanding

1 Score 3 NC.8.EE.7

Which is equal to $3(x + 4)$?

(A) 3x + 4 (B) x + 12

(C) 3x + 12 (D) 3x + 7

2 Score 3 NC.8.EE.7

Combine like terms: $5x + 2 + 3x$

(A) $8x^2$ (B) 15x + 2

(C) 10x (D) 8x + 2

3 Score 3 NC.8.EE.7

Solve: $4(x - 2) = 20$

Answer: _____

3.3 Variables on both sides NC.8.EE.7

When x is on **both sides**, move the x-terms to one side first (subtract the smaller x-term from both sides). Then solve the two-step equation that is left.

Like terms

5x and 3x are like terms, so 5x − 3x = 2x

Both sides

the left of the = sign and the right of the = sign

The 5-step plan:

1. Distribute (if there are parentheses).
2. Combine like terms on each side.
3. Move the x-terms to one side.
4. Move the numbers to the other side.
5. Divide to get x alone.

Watch out: Subtract the x-term from **both** sides, not just the side it is on.

1 $5x + 2 = 3x + 10$ → Subtract 3x
$-3x \quad -3x$

2 $2x + 2 = 10$ → Subtract 2
$-2 \quad -2$

3 $\frac{2x}{2} = \frac{8}{2}$ → Divide by 2

4 $x = 4$

Walk through: Solve $7x + 3 = 4x + 15$

1 x is on both sides. What move puts the x's together?

2 Undo the + 3.

3 Get x alone.

Check your understanding

1 Score 3 NC.8.EE.7

What is a good **first** move for $6x - 1 = 2x + 11$?

(A) Subtract 2x from both sides

(B) Divide both sides by 6

(C) Add 2x to both sides

(D) Subtract 6x from the right side only

2 Score 3 NC.8.EE.7

Solve: $6x - 1 = 2x + 11$

Answer: _____

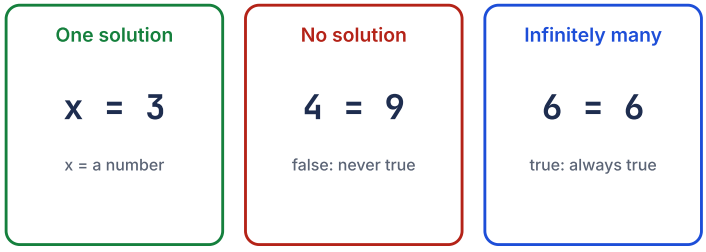
3.4 One, none, or infinitely many solutions NC.8.EE.7a

Solve as usual. If the x's cancel, look at what is left. **False** (like 4 = 9) means **no solution**. **True** (like 6 = 6) means **infinitely many solutions**. If you get x = a number, there is **one solution**.

No solution
no number makes the equation true

Infinitely many solutions
every number makes the equation true

One solution
exactly one number works, like x = 3



$2x + 4 = 2x + 9 \rightarrow 4 = 9$ (false) → no solution

$3(x + 2) = 3x + 6 \rightarrow 6 = 6$ (true) → infinitely many

Watch out: If the x's cancel, you are not stuck. The last line tells you the answer.

Walk through: How many solutions? 3(x + 2) = 3x + 6

1. First move?

2. Subtract 3x from both sides. What is left?

3. 6 = 6 is true. How many solutions?

Check your understanding

1 Score 3 NC.8.EE.7a

How many solutions does **2x + 4 = 2x + 9** have?

(A) No solution
(B) One solution
(C) Infinitely many solutions

2 Score 3 NC.8.EE.7a

A last line says **5 = 5**. How many solutions?

(A) Infinitely many solutions
(B) One solution
(C) No solution

3 Score 3 NC.8.EE.7a

How many solutions does **4x + 1 = 2x + 7** have?

(A) One solution
(B) No solution
(C) Infinitely many solutions

3.5 Inequalities: solve and graph NC.8.EE.7

Solve an inequality like an equation, with **one special rule**: when you **multiply or divide by a negative number, flip the sign**. Then graph it on a number line.

< and >

less than, greater than: open circle (the number is not included)

≤ and ≥

at most, at least: closed circle (the number is included)

Flip

turn the sign around: < becomes >, ≤ becomes ≥



x > 2: open circle at 2, shade right

Watch out: Only flip when you **multiply or divide by a negative**. Adding or subtracting never flips.

$-2x < 6$

$\div(-2) \quad \div(-2)$

$x > -3$



Walk through: Solve 2x + 1 ≥ 7 and graph it

1. Undo the + 1.

2. Undo the × 2. Do we flip?

3. Open or closed circle at 3?

4. Which way do we shade?

Check your understanding

1 Score 3 NC.8.EE.7

Solve: $x + 4 > 9$

(A) $x \geq 5$ (B) $x > 5$

(C) $x < 5$ (D) $x > 13$

2 Score 3 NC.8.EE.7

Solve: $-2x < 8$

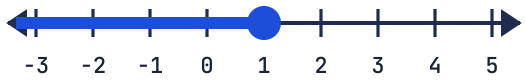
(A) $x > -4$ (B) $x < -4$

(C) $x < -16$ (D) $x > 4$

3 Score 3 NC.8.EE.7

Which graph shows $x \leq 1$?

A



B



C

